

Miraculous Owonubi

Senior Rust Engineer · Distributed Systems, P2P Networking, Protocol

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Highlights

- **NEAR Protocol** (core engineer): creator and principal author of its Rust JSON-RPC client, and 80+ merged PRs across nearcore itself, NEAR's core protocol client.
- **Calimero** (engineering lead): built the protocol that became the company's direction, designed both generations, primary reviewer across the codebase.
- Shipped, adopted open source: **freyr** at 2.3k stars and 42k+ Docker pulls, plus upstream work in tickrs, Apache Arrow, rustyline, and dialoguer.

Experience

Independent Engineering & Consulting

September 2025 – Present

Rust Systems & Protocol Engineering

Remote

- **Real-time bidding (adtech)**: profiled the hot path to cut sustained tail-latency spikes from roughly 250ms to within a hard 150ms budget at tens of thousands of RPS, then built the analytics pipeline that turns the bid-stream into signals (competitor-landscape, security, forensics).
- **theia** (Rust, github.com/theia-hq): a public-key-addressed overlay and services layer I architected and built. Reach any machine by its ed25519 key, then gate each service on a capability rooted in that key: attenuable, verified offline, revocable via a node-checked denylist. Services (keyless SSH, HTTP fetch, BLAKE3-verified transfer) run over iroh or a UDP transport I wrote from scratch to learn one, behind one CLI (swoosh), backed by 300+ tests.
- **Private EVM L2**: built the sequencing and execution engine in Rust: an actor-model core and a purpose-built zero-copy storage layer with snapshot-isolated reads and atomic block-commit, on revm and alloy.

Calimero Network

September 2023 – August 2025

Senior Software Engineer, then Engineering Lead

Remote

- Calimero is a peer-to-peer app platform: apps run as WASM on each user's own node, and a group's shared state syncs between nodes over CRDTs, end-to-end encrypted. I was brought in as the systems engineer for its hardest problems; when the original private-sharding direction stalled, the protocol I prototyped won the internal hackathon and became that platform.
- Architected and built that protocol's stack end to end: the node-to-node wire format, networking (rust-libp2p, hole-punching and relay), CRDT-state foundations, recursive state sync between nodes, an actor-model core, a purpose-built zero-copy storage layer, a Wasmer runtime with a swappable engine, and the guest-app SDK (proc-macros plus FFI).
- Designed both generations of the protocol, a transactional proof-of-concept then an eventually-consistent redesign, and its end-to-end encryption for dynamic groups; authored the adopted spec paper and a paper on the encryption.
- As Engineering Lead, grew a team mostly new to Rust through hands-on knowledge-transfer, and hired and developed several engineers; primary reviewer across protocol proposals and the codebase.

NEAR (Pagoda, Inc.)

March 2021 – June 2023

Senior Software Engineer, Core & Developer Tools (December 2021 – June 2023)

Remote

- 80+ merged PRs across nearcore, NEAR's core protocol client, and its Rust tooling; creator and principal author of the Rust NEAR JSON-RPC client, which seeded the sandboxed node testing environment, the Rust NEAR CLI, and the indexer for the official NEAR explorer.
- Introduced contract ABIs to NEAR: authored the NearSchema derive in near-sdk-rs and contributed to cargo-near; discovered and documented exploits in the NEAR BOS platform that fed into a working group for reinventing its virtual machine.
- Cut CI and Docker build times by up to 82%, most notably for the FastAuth account-recovery node tests.

Senior Software Engineer, Node Interfaces (March 2021 – December 2021)

- Led the standardization of a strictly-typed AccountId, drove workspace crates to independence for crates.io, and built the tooling (themis) and CI pipelines for crate compliance, fuzz, and simulation testing.

Open Source & Freelance

January 2016 – Present

Software Engineer

Remote

- **freyr** (Node.js): an open-source music-downloader CLI (Spotify, Apple Music, Deezer) with metadata and library organization. 2.3k+ stars, 42k+ DockerHub pulls.
- **cargo-workspaces** (Rust): a maintained fork with substantial patches for nearcore's workspace complexity; **xget** (Node.js): a concurrent chunked web-content downloader.
- **Upstream**: merged contributions to tickrs (1.7k-star stock TUI, 17 PRs), Apache Arrow's object_store, rustyline, and dialoguer.

Skills

Languages: Rust; Python, JavaScript, TypeScript (proficient); C/C++, Go (reading fluency).

Systems & networking: P2P and mesh networking, libp2p, iroh, custom transports, QUIC, NAT traversal, async Rust (tokio); storage engines (RocksDB, LMDB); operating systems, Linux internals, virtualization, containers.

Distributed systems & protocol: consensus and protocol design, Ethereum execution (revm, alloy), WebAssembly (WASM) runtimes, applied cryptography and capability security.

Craft: clean API design (libraries, REST, JSON-RPC), CLI and TUI, CI/CD, debugging, profiling, reverse engineering, exhaustive testing (unit, integration, fuzz).

Education

Redeemer's University, Osun, Nigeria

July 2018 – May 2021

B.Sc., Computer Science

Operating Systems, Compiler Architecture, Language Development, Databases, Algorithms.

Interests: distributed systems, applied cryptography, zero-knowledge proofs, information theory, fault tolerance.